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$$\begin{aligned}f(x) &= \frac{\sqrt{x+7}-3}{x-2} \\ \lim_{x \rightarrow 2} \frac{\sqrt{x+7}-3}{x-2} &= \frac{\sqrt{9}-3}{0} = \frac{0}{0} = \infty \\ &= \lim_{x \rightarrow 2} \frac{(\sqrt{x+7}-3)(\sqrt{x+7}+3)}{(x-2)(\sqrt{x+7}+3)} \\ &= \lim_{x \rightarrow 2} \frac{x+7-9}{(x-2)(\sqrt{x+7}+3)} \\ &= \lim_{x \rightarrow 2} \frac{1}{\sqrt{x+7}+3} = \frac{1}{6}\end{aligned}$$

References